

DANIEL J. VARON

Curriculum Vitae | May 2026

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EDUCATION

Ph.D., Atmospheric Chemistry , Harvard University	2015 – 2020
M.Sc., Applied Mathematics	
Secondary field in Computational Science & Engineering	
B.A., English Literature , McGill University	2010 – 2014
B.Sc., Physics , McGill University	2009 – 2014

PROFESSIONAL EXPERIENCE

Assistant Professor , Massachusetts Institute of Technology	2025 –
Boeing Career Development Professor of Aeronautics and Astronautics	
Department of Aeronautics and Astronautics	
Institute for Data, Systems, and Society	
Research Associate , Harvard University	2023 – 2025
School of Engineering and Applied Sciences	
Visiting Postdoctoral Research Associate , Princeton University	2021 – 2023
School of Public and International Affairs	
Postdoctoral Research Fellow , Harvard University	2020 – 2023
School of Engineering and Applied Sciences	

PUBLICATIONS (*SUBMITTED)

h-index = 30, total citations = 4727 (as of May 2026 on [Google Scholar](https://scholar.google.com/))

- *66. Gauld, K. D., **Varon, D. J.**, Thorpe, A. K., and Green, R. O.: Quantifying Nitrogen Oxide Point Sources with the EMIT Satellite Imaging Spectrometer, submitted to *Geophys. Res. Lett.*, 2026.
- *65. Balasus, N., Cusworth, D. H., Kim, J., **Varon, D. J.**, Miller, C. E., and Duren, R. M.: Mapping ammonia emission plumes using shortwave infrared imaging spectroscopy, in review [preprint], <https://doi.org/10.48550/arXiv.2602.12488>, 2025.
- *64. Hancock, S. E., Jacob, D. J., Jimenez, R., Ardila, A., Morales-Rincon, L., Rojas, N., Estrada, L. A., Balasus, N., East, J. D., Sulprizio, M. P., Wang, X., France, J. L., Potyk, L., Penn, E., Chen, Z., **Varon, D. J.**, Frankenberg, C., Baranski, M., Calcan, A., and Parker, R. J.: Applying Satellite Observations to Improve Bottom-Up National Emission Inventories for Methane: Application to Colombia, EGU sphere [preprint] *Atmos. Chem. Phys.*, <https://doi.org/10.5194/egusphere-2025-5478>, 2026.
- *63. Estrada, L. A., Jacob, D. J., He, M., East, J. D., **Varon, D. J.**, Balasus, N., Hancock, S. E., Sulprizio, M., Bowman, K. W., Worden, J. R., Reidy, E., and Runkle, B. R. K.: Quantifying national, state, and oil/gas field methane emissions and trends in the U.S. (2019-2024) through high resolution inversion of satellite observations, [preprint], <https://doi.org/10.5194/egusphere-2026-655>, 2026.
- *62. He, T.-L., **Varon, D. J.**, Kondragunta, S., Ren, X., Cohen, M. D., Carroll, B. J., Malarich, M., Peischl, J., de Jong, T. A., Gerritsen, J., Maasakkers, J. D., Cusworth, D. H., Duren, R. M., Brown, S. S., Warneke, C., Sweeney, C., Stratton, P., Brewer, A., Baidar, S., and Levin, E. J. T.: Coordinated satellite, aircraft, and ground-based observations of a large transient methane release, [preprint], <https://doi.org/10.31223/X54R07>, 2026.

- *61. Worden, J. R., Pandey, S., Nesser, H. O., Bowman, K. W., Harkins, C., Lyi, C., Maasakkers, J. D., Gordon, D., Jacob, D. J., Estrada, L. A., **Varon, D. J.**, East, J. D., Schmeisser, L., and Qu, Z.: Top-Down Benchmark of U.S. Methane Inventories Reveals Regional Discrepancies in Activity-Based Estimates, [preprint] *Atmos. Chem. Phys.*, <https://doi.org/10.5194/egusphere-2026-313>, 2026.
- *60. Lama, S., Maasakkers, J. D., Zhang, X., Girard, M., Dutt, S., Nandgaonkar, S., **Varon, D. J.**, Sulprizio, M. P., Estrada, L. A., Balasus, N., Parker, R. J., Terao, Y., and Aben, I.: Evaluating national, state, and urban Indian methane emissions using satellites, *EGUsphere* [preprint], <https://doi.org/10.5194/egusphere-2025-6528>, 2026.
- *59. Chen, Z., Jacob, D. J., Lin, H., Balasus, N., Hancock, S. E., Estrada, L. A., East, J. D., Zhang, Y., Wang, X., He, M., **Varon, D. J.**, and Lunt, M. F.: Vegetation-driven steady increases in tropical wetland methane emissions over the past two decades, in review, 2026.
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- *57. Bruno, J. H., Jacob, D. J., Wang, X., Sulprizio, M. P., Estrada, L. A., **Varon, D. J.**, Wofsy, S. C., Omara, M., and Gautam, R.: Integrating MethaneAIR aircraft and TROPOMI satellite observations in the Integrated Methane Inversion (IMI) to optimize methane emissions, [preprint], <https://doi.org/10.5194/egusphere-2025-4626>, 2026.
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32. Bruno, J. H., Jervis, D., **Varon, D. J.**, and Jacob, D. J.: U-Plume: automated algorithm for plume detection and source quantification by satellite point-source imagers, *Atmos. Meas. Tech.*, 17, 2625–2636, <https://doi.org/10.5194/amt-17-2625-2024>, 2024.
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29. Schuit, B. J., Maasakkers, J. D., Bijl, P., Mahapatra, G., van den Berg, A.-W., Pandey, S., Lorente, A., Borsdorff, T., Houweling, S., **Varon, D. J.**, McKeever, J., Jervis, D., Girard, M., Irakulis-Loitxate, I., Gorroño, J., Guanter, L., Cusworth, D. H., and Aben, I.: Automated detection and monitoring of methane super-emitters using satellite data, *Atmos. Chem. Phys.*, 23, 9071–9098, <https://doi.org/10.5194/acp-23-9071-2023>, 2023.
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24. **Varon, D. J.**, Jacob, D. J., Hmiel, B., Gautam, R., Lyon, D. R., Omara, M., Sulprizio, M., Shen, L., Pendergrass, D., Nesser, H., Qu, Z., Barkley, Z. R., Miles, N. L., Richardson, S. J., Davis, K. J., Pandey, S., Lu, X., Lorente, A., Borsdorff, T., Maasakkers, J. D., and Aben, I.: Continuous weekly monitoring of methane emissions from the Permian Basin by inversion of TROPOMI satellite observations, *Atmos.*

Selected as Highlight Paper

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22. Lu, X., Jacob, D. J., Zhang, Y., Shen, L., Sulprizio, M. P., Maasakkers, J. D., **Varon, D. J.**, Qu, Z., Chen, Z., Hmiel, B., Parker, R. J., Boesch, H., Wang, H., He, C., and Fan, S.: Observation-derived 2010–2019 trends in methane emissions and intensities from US oil and gas fields tied to activity metrics, *Proc. Natl. Acad. Sci.*, <https://doi.org/10.1073/pnas.2217900120> 2023.
21. Gorroño, J., **Varon, D. J.**, Irakulis-Loitxate, I., and Guanter, L.: Understanding the potential of Sentinel-2 for monitoring methane point emissions, *Atmos. Meas. Tech.*, 16, 89–107, <https://doi.org/10.5194/amt-16-89-2023>, 2023.
20. Zhang, Z., Sherwin, E. D., **Varon, D. J.**, and Brandt, A. R.: Detecting and quantifying methane emissions from oil and gas production: algorithm development with ground-truth calibration based on Sentinel-2 satellite imagery, *Atmos. Meas. Tech.*, 15, 7155–7169, <https://doi.org/10.5194/amt-15-7155-2022>, 2022.
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18. Chen, Z., Jacob, D. J., Nesser, H., Sulprizio, M. P., Lorente, A., **Varon, D. J.**, Lu, X., Shen, L., Qu, Z., Penn, E., and Yu, X.: Methane emissions from China: a high-resolution inversion of TROPOMI satellite observations, *Atmos. Chem. Phys.*, 22, 10809–10826, <https://doi.org/10.5194/acp-22-10809-2022>, 2022.
17. Qu, Z., Jacob, D. J., Zhang, Y., Shen, L., **Varon, D. J.**, Lu, X., Scarpelli, T., Bloom, A., Worden, J., and Parker, R. J.: Attribution of the 2020 surge in atmospheric methane by inverse analysis of GOSAT observations, *Environ. Res. Lett.*, 17, 9, <https://doi.org/10.1088/1748-9326/ac8754>, 2022.
16. Maasakkers, J. D., **Varon, D. J.**, Elfarsdóttir, A., McKeever, J., Jervis, D., Mahapatra, G., Pandey, S., Lorente, A., Borsdorff, T., Foorthuis, L. R., Schuit, B. J., Tol, P., van Kempen, T. A., van Hees, R., and Aben, I.: Using satellites to uncover large methane emissions from landfills, *Sci. Adv.*, 8, 32, <https://doi.org/10.1126/sciadv.abn9683>, 2022.
15. Jacob, D. J., **Varon, D. J.**, Cusworth, D. H., Dennison, P. E., Frankenberg, C., Gautam, R., Guanter, L., Kelley, J., McKeever, J., Ott, L. E., Poulter, B., Qu, Z., Thorpe, A. K., Worden, J. R., and Duren, R. M.: Quantifying methane emissions from the global scale down to point sources using satellite observations of atmospheric methane, *Atmos. Chem. Phys.*, 22, 9617–9646, <https://doi.org/10.5194/acp-22-9617-2022>, 2022.
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Among AMT’s most downloaded: https://amt.copernicus.org/most_downloaded.html.
Selected as Highlight Paper
8. Jervis, D., McKeever, J., Durak, B. O. A., Sloan, J. J., Gains, D., **Varon, D. J.**, Ramier, A., Strupler, M., and Tarrant, E.: The GHGSat-D Imaging Spectrometer. *Atmos. Meas. Tech. Discuss.*, 14, 2127–2140, <https://doi.org/10.5194/amt-14-2127-2021>, 2021.
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5. Zhang, Y., Gautam, R., Pandey, S., Omara, M., Maasakkers, J. D., Sadavarte, P., Lyon, D., Nesser, H., Sulprizio, M. P., **Varon, D. J.**, Zhang, R., Houweling, S., Zavala-Araiza, D., Alvarez, R. A., Lorente, A., Hamburg, S. P., Aben, I., and Jacob, D. J.: Quantifying methane emissions from the largest oil producing basin in the U.S. from space. *Science Advances*, 6, 17, <https://www.science.org/doi/10.1126/sciadv.aaz51>, 2020.
4. Cusworth, D. H., Jacob, **D. J.**, **Varon, D. J.**, Chan Miller, C., Liu, X., Chance, K., Thorpe, A. K., Duren, R. M., Miller, C. E., Thompson, D. R., Frankenberg, C., Guanter, L., and Randles, C. A.: Potential of next-generation imaging spectrometers to detect and quantify methane point sources from space, *Atmos. Meas. Tech.*, 12, 5655–5668, <https://doi.org/10.5194/amt-12-5655-2019>, 2019.
3. **Varon, D. J.**, McKeever, J., Jervis, D., Maasakkers, J. D., Pandey, S., Houweling, S., Aben, I., Scarpelli, T., and Jacob, D. J.: Satellite discovery of anomalously large methane point sources from oil/gas production. *Geophys. Res. Lett.*, 46, 22, <https://doi.org/10.1029/2019GL083798>, 2019.
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2. **Varon, D. J.**, Jacob, D. J., McKeever, J., Jervis, D., Durak, B. O. A., Xia, Y., and Huang, Y.: Quantifying methane point sources from fine-scale satellite observations of atmospheric methane plumes. *Atmos. Meas. Tech.*, 11, 5673–5686, <https://doi.org/10.5194/amt-11-5673-2018>, 2018.
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MENTORING

Postdocs

- Tailong He (Harvard, MIT), 2024– . GOES-ABI methane retrievals.

Graduate students

- Krishma Mehta Rajesh (PhD, MIT), 2025–
- Vassiliki Mancoridis (PhD, MIT), 2025–
- Kevin Gauld (PhD, MIT), 2025–
- François Martin-Monier (MSc, ETH Zürich), 2023. ML-based Sentinel-2 methane detection.
- Marc Watine (MSc, ETH Zürich), 2023. Geostationary satellite methane retrievals.

Undergraduate students

- Jayden Graham (MIT), 2026. Landsat NO₂ retrievals.
- Asael (Acosta), 2025. Geostationary satellite methane retrievals.
- Khue Phan (MIT), 2025. Ocean color satellite DMS detection.
- Israel Garcia (MIT), 2025. Ocean color satellite DMS detection.
- Chevaugn Campbell (Kenyon College), 2022. Landsat methane retrievals.
- Daniel Shen (Harvard University), 2021. Sentinel-2 methane retrievals.

SERVICE

Professional Service

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|-------|--|
| 2025– | Model Scientist, Integrated Methane Inversion (IMI) |
| 2025– | Co-Nested-Model Scientist, GEOS-Chem |
| 2024– | Scientific Advisory Board, Methane Emissions Detection Using Satellites Assessment (MEDUSA) project (SRON Space Research Organisation Netherlands) |
| 2023– | Associate Editor, <i>Atmospheric Measurement Techniques</i> |